

Assignment: IPv4 Subnetting & VLSM

Q1. Understanding Subnetting (2 Marks)

A company has been assigned the network 192.168.10.0/24. Explain why subnetting is required in computer networks. Mention at least three advantages of subnetting. If the company divides this network into 4 equal-sized subnets, determine: 1.Subnet mask. 2.Number of subnets. 3.Number of usable hosts per subnet.

(Show all necessary calculations.)

Q2. Subnetting Problem (3 Marks)

An organization owns the network 172.16.0.0/16. It wants to create 16 equal-sized subnets. Determine the following:

1. New subnet mask (CIDR notation and dotted decimal)
2. Number of subnet bits borrowed
3. Number of usable hosts per subnet
4. List the first four valid subnet addresses
5. Find the: Network Address, Broadcast Address, First Usable Host, Last Usable Host for the third subnet.

(Show all necessary calculations.)

Q3. Variable Length Subnet Mask (VLSM) Design (3 Marks)

A university has been assigned the network 192.168.100.0/24. The university needs separate networks for the following departments:

Department	Required Hosts
Computer Science	100
Business	50
EEE	25
Administration	10

Using VLSM, design an efficient addressing scheme. For each department provide: Network Address, Subnet Mask (/ notation and dotted decimal), First Usable Host, Last Usable Host, Broadcast Address. Explain why VLSM is more efficient than Fixed-Length Subnet Masking (FLSM) in this scenario.

(Show all necessary calculations.)

Q4. Critical Thinking (2 Marks)

Two network engineers propose different addressing methods for the same organization.
Engineer A: Divide the network into equal-sized subnets using traditional subnetting (FLSM).
Engineer B: Use Variable Length Subnet Masking (VLSM).

Assume the organization has departments requiring:

1. 120 hosts
2. 60 hosts
3. 30 hosts
4. 10 hosts

a) Which approach would waste more IP addresses? Explain with reasoning.

b) Recommend the better solution and justify your answer based on: Efficient IP address utilization, Scalability and Ease of future expansion

(Show all necessary calculations.)